

MH1201 Linear Algebra II

Midterm Exam (Solutions)

Practice Paper – 1

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Subspace

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Question 1 (Single Linear Constraint in $\mathbb{R}^3$) (8 marks)

Problem

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x - 2y + 3z = 0\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Kernel of a linear functional

Define

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad T(x, y, z) = (1, -2, 3) \cdot (x, y, z).$$

Then

$$T(x, y, z) = x - 2y + 3z.$$

Since the dot product with a fixed vector defines a linear functional, T is linear. Hence

$$W = \ker(T).$$

Since the kernel of a linear map is a subspace, W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Method 2: Parametrisation as a span

Let $y = s$ and $z = t$, where $s, t \in \mathbb{R}$. Then

$$x - 2y + 3z = 0 \implies x = 2s - 3t.$$

Thus every vector in W has the form

$$(x, y, z) = (2s - 3t, s, t) = s(2, 1, 0) + t(-3, 0, 1).$$

Therefore

$$W = \text{span}\{(2, 1, 0), (-3, 0, 1)\}.$$

Since a span is always a subspace, W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Question 2 (Symmetric 2×2 Matrices)

(8 marks)

Problem

Let

$$W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid A^T = A\}.$$

Determine whether W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Solution

Method 1: Closure under linear combinations

Let $A, B \in W$. Then

$$A^T = A, \quad B^T = B.$$

For any $a, b \in \mathbb{R}$,

$$(aA + bB)^T = aA^T + bB^T = aA + bB.$$

Hence $aA + bB \in W$. Therefore W is closed under linear combinations, so W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Method 2: Explicit span description

Let

$$A = \begin{pmatrix} x & y \\ z & t \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}).$$

The condition $A^T = A$ is equivalent to

$$y = z.$$

Hence every element of W has the form

$$\begin{pmatrix} x & y \\ y & t \end{pmatrix} = x \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + y \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + t \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus

$$W = \text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

Therefore W is a subspace.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Question 3 (Polynomials Vanishing at $x = 1$) (8 marks)

Problem

Let

$$W = \{p(x) \in \mathcal{P}_2(\mathbb{R}) \mid p(1) = 0\}.$$

Determine whether W is a subspace of $\mathcal{P}_2(\mathbb{R})$.

Solution

Method 1: Kernel of the evaluation map

Define

$$T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathbb{R}, \quad T(p) = p(1).$$

Evaluation at a fixed point defines a linear functional on $\mathcal{P}_2(\mathbb{R})$. Hence T is linear. Therefore

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $\mathcal{P}_2(\mathbb{R})$.

$W \leq \mathcal{P}_2(\mathbb{R})$

Method 2: Factorisation

Let $p \in W$. Then $p(1) = 0$, so $x = 1$ is a root of $p(x)$. Hence $x - 1$ is a factor of $p(x)$, and since p has degree at most 2, there exist $a, b \in \mathbb{R}$ such that

$$p(x) = (x - 1)(ax + b).$$

Expanding,

$$p(x) = ax(x - 1) + b(x - 1).$$

Thus

$$W = \text{span}\{x(x - 1), x - 1\}.$$

Therefore W is a subspace of $\mathcal{P}_2(\mathbb{R})$.

$W \leq \mathcal{P}_2(\mathbb{R})$

Question 4 (Singular 2×2 Matrices)

(8 marks)

Problem

Let

$$W = \{A \in M_{2 \times 2}(\mathbb{R}) \mid \det(A) = 0\}.$$

Determine whether W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Solution

Method 1: Failure of closure under addition

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then

$$\det(A) = 0, \quad \det(B) = 0,$$

so $A, B \in W$. However,

$$A + B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I,$$

and

$$\det(I) = 1 \neq 0.$$

Thus $A + B \notin W$. Hence W is not closed under addition, so W is not a subspace.

$$\boxed{W \not\subseteq M_{2 \times 2}(\mathbb{R})}$$

Method 2: Union of one-dimensional subspaces

A 2×2 matrix is singular if and only if its columns are linearly dependent. In particular, every matrix of the form

$$\begin{pmatrix} a & \lambda a \\ b & \lambda b \end{pmatrix}$$

is singular for any fixed $\lambda \in \mathbb{R}$.

For each fixed λ , define

$$U_\lambda = \left\{ \begin{pmatrix} a & \lambda a \\ b & \lambda b \end{pmatrix} : a, b \in \mathbb{R} \right\}.$$

Each U_λ is a subspace of $M_{2 \times 2}(\mathbb{R})$, and W is the union of many such subspaces together with the subspace of matrices whose second column is zero. Since W contains, for example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

which lie in different such subspaces, but their sum is invertible, W cannot itself be a subspace.

Therefore

$$\boxed{W \not\subseteq M_{2 \times 2}(\mathbb{R})}.$$

Question 5 (One Linear Relation in $\mathbb{R}^3$)

(8 marks)

Problem

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x = 2y\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Parametrisation

A vector $(x, y, z) \in W$ satisfies $x = 2y$. Let $y = s$ and $z = t$. Then

$$(x, y, z) = (2s, s, t) = s(2, 1, 0) + t(0, 0, 1).$$

Hence

$$W = \text{span}\{(2, 1, 0), (0, 0, 1)\}.$$

Therefore W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Method 2: Kernel trick

Define

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad T(x, y, z) = (-1, 2, 0) \cdot (x, y, z).$$

Then

$$T(x, y, z) = -x + 2y.$$

Since the dot product with a fixed vector defines a linear functional, T is linear. Also,

$$x = 2y \iff -x + 2y = 0 \iff T(x, y, z) = 0.$$

Thus

$$W = \ker(T).$$

Since kernels are subspaces, W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Question 6 (Affine Plane in $\mathbb{R}^3$)

(8 marks)

Problem

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y + z = 1\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Zero vector test

If W were a subspace, then it would contain the zero vector $(0, 0, 0)$. However,

$$0 + 2 \cdot 0 + 0 = 0 \neq 1.$$

Thus

$$(0, 0, 0) \notin W.$$

Therefore W is not a subspace of $\mathbb{R}^3$.

$$\boxed{W \not\subseteq \mathbb{R}^3}$$

Method 2: Affine translate of a kernel

Define

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad T(x, y, z) = (1, 2, 1) \cdot (x, y, z).$$

Then

$$T(x, y, z) = x + 2y + z,$$

so

$$W = \{v \in \mathbb{R}^3 \mid T(v) = 1\}.$$

Let

$$U = \ker(T) = \{v \in \mathbb{R}^3 \mid T(v) = 0\}.$$

Choose $v_0 = (1, 0, 0)$. Then $T(v_0) = 1$, so for any $v \in W$,

$$T(v - v_0) = T(v) - T(v_0) = 1 - 1 = 0,$$

hence $v - v_0 \in U$. Therefore

$$W = v_0 + U.$$

So W is an affine translate of the subspace U . Since $v_0 \neq 0$, this affine set does not pass through the origin, and therefore it is not a subspace.

$$\boxed{W \not\subseteq \mathbb{R}^3}$$

Question 7 (Trace-Zero Matrices)

(8 marks)

Problem

Let

$$W = \{A \in M_{n \times n}(\mathbb{R}) \mid \operatorname{tr}(A) = 0\}.$$

Determine whether W is a subspace of $M_{n \times n}(\mathbb{R})$.

Solution

Method 1: Linear combination test

Let $A, B \in W$. Then

$$\operatorname{tr}(A) = 0, \quad \operatorname{tr}(B) = 0.$$

For any $a, b \in \mathbb{R}$,

$$\operatorname{tr}(aA + bB) = a \operatorname{tr}(A) + b \operatorname{tr}(B) = 0.$$

Hence $aA + bB \in W$. Therefore W is a subspace of $M_{n \times n}(\mathbb{R})$.

$$\boxed{W \leq M_{n \times n}(\mathbb{R})}$$

Method 2: Kernel of the trace map

Define

$$T : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}, \quad T(A) = \operatorname{tr}(A).$$

Since

$$\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B), \quad \operatorname{tr}(cA) = c \operatorname{tr}(A),$$

the map T is linear. Thus

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $M_{n \times n}(\mathbb{R})$.

$$\boxed{W \leq M_{n \times n}(\mathbb{R})}$$

Question 8 (Intersection of Two Subspaces) (10 marks)

Problem

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0, x - y = 0\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Intersection property

Define

$$U_1 = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\},$$

and

$$U_2 = \{(x, y, z) \in \mathbb{R}^3 \mid x - y = 0\}.$$

Each of U_1 and U_2 is a subspace, since each is the solution set of a homogeneous linear equation. Moreover,

$$W = U_1 \cap U_2.$$

The intersection of subspaces is always a subspace. Therefore W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Method 2: Explicit parametrisation

From

$$x - y = 0$$

we get

$$x = y.$$

Substitute this into

$$x + y + z = 0.$$

Then

$$y + y + z = 0 \implies 2y + z = 0 \implies z = -2y.$$

Hence every vector in W has the form

$$(x, y, z) = (y, y, -2y) = y(1, 1, -2).$$

Therefore

$$W = \text{span}\{(1, 1, -2)\},$$

which is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Question 9 (Smallest Subspace Containing a Set) (10 marks)

Problem

Let

$$S = \{(1, 0, 1), (2, 1, 3), (3, 1, 4)\} \subset \mathbb{R}^3.$$

Let W be the smallest subspace of $\mathbb{R}^3$ containing S .

Determine W .

Solution

Method 1: Use the definition of span

The smallest subspace containing a set is its span. Hence

$$W = \text{span}(S) = \text{span}\{(1, 0, 1), (2, 1, 3), (3, 1, 4)\}.$$

Observe that

$$(3, 1, 4) = (1, 0, 1) + (2, 1, 3).$$

Thus the third vector is redundant, so

$$W = \text{span}\{(1, 0, 1), (2, 1, 3)\}.$$

Therefore

$$\boxed{W = \text{span}\{(1, 0, 1), (2, 1, 3)\}}.$$

Method 2: Minimality argument

Let

$$U = \text{span}\{(1, 0, 1), (2, 1, 3)\}.$$

Then U is a subspace of $\mathbb{R}^3$. Also,

$$(1, 0, 1) \in U, \quad (2, 1, 3) \in U,$$

and since U is closed under addition,

$$(3, 1, 4) = (1, 0, 1) + (2, 1, 3) \in U.$$

Hence $S \subseteq U$. Therefore U is a subspace containing S .

Now let V be any subspace containing S . Then V contains $(1, 0, 1)$ and $(2, 1, 3)$, so by closure under linear combinations, V must contain

$$\text{span}\{(1, 0, 1), (2, 1, 3)\} = U.$$

Thus every subspace containing S contains U . Therefore U is the smallest subspace containing S , and so

$$\boxed{W = \text{span}\{(1, 0, 1), (2, 1, 3)\}}.$$

Question 10 (Derivative Constraint on $\mathcal{P}_3(\mathbb{R})$) (10 marks)

Problem

Let

$$W = \{p(x) \in \mathcal{P}_3(\mathbb{R}) \mid p'(1) = 0\}.$$

Determine whether W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

Solution

Method 1: Kernel of a linear map

Define

$$T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathbb{R}, \quad T(p) = p'(1).$$

We first verify that T is linear. For $p, q \in \mathcal{P}_3(\mathbb{R})$ and $a, b \in \mathbb{R}$,

$$T(ap + bq) = (ap + bq)'(1) = (ap' + bq')(1) = ap'(1) + bq'(1) = aT(p) + bT(q).$$

Hence T is linear. Therefore

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

$W \leq \mathcal{P}_3(\mathbb{R})$

Method 2: Explicit general form

Let

$$p(x) = ax^3 + bx^2 + cx + d \in \mathcal{P}_3(\mathbb{R}).$$

Then

$$p'(x) = 3ax^2 + 2bx + c,$$

so

$$p'(1) = 3a + 2b + c.$$

Thus $p \in W$ if and only if

$$3a + 2b + c = 0.$$

Solve for c :

$$c = -3a - 2b.$$

Hence

$$p(x) = ax^3 + bx^2 - (3a + 2b)x + d = a(x^3 - 3x) + b(x^2 - 2x) + d.$$

Therefore

$$W = \text{span}\{x^3 - 3x, x^2 - 2x, 1\}.$$

Since a span is a subspace, W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

$W \leq \mathcal{P}_3(\mathbb{R})$

Question 11 (Row-Sum-Zero Matrices)

(10 marks)

Problem

Let

$$W = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid a + b = 0, c + d = 0 \right\}.$$

Determine whether W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Solution

Method 1: Linear combination test

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \in W.$$

Then

$$a + b = 0, \quad c + d = 0, \quad e + f = 0, \quad g + h = 0.$$

For any $\lambda, \mu \in \mathbb{R}$,

$$\lambda A + \mu B = \begin{pmatrix} \lambda a + \mu e & \lambda b + \mu f \\ \lambda c + \mu g & \lambda d + \mu h \end{pmatrix}.$$

Its first row sum is

$$(\lambda a + \mu e) + (\lambda b + \mu f) = \lambda(a + b) + \mu(e + f) = 0,$$

and similarly its second row sum is

$$(\lambda c + \mu g) + (\lambda d + \mu h) = \lambda(c + d) + \mu(g + h) = 0.$$

Hence $\lambda A + \mu B \in W$. Therefore W is a subspace.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Method 2: Kernel of a linear map

Define

$$T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = (a + b, c + d).$$

This map is linear, since each component is linear in the entries of the matrix. Moreover,

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Method 3: Explicit span description

The conditions

$$a + b = 0, \quad c + d = 0$$

give

$$b = -a, \quad d = -c.$$

Hence every matrix in W has the form

$$\begin{pmatrix} a & -a \\ c & -c \end{pmatrix} = a \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} + c \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix}.$$

Thus

$$W = \text{span} \left\{ \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \right\}.$$

Therefore W is a subspace.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Question 12 (Affine Hyperplane in $\mathbb{R}^3$)

(10 marks)

Problem

Let

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x - y + 2z = 3\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Zero vector test

If W were a subspace, then it would contain $(0, 0, 0)$. However,

$$0 - 0 + 2 \cdot 0 = 0 \neq 3.$$

Thus $(0, 0, 0) \notin W$. Hence W is not a subspace.

$$\boxed{W \not\leq \mathbb{R}^3}$$

Method 2: Affine translate of a kernel

Define

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}, \quad T(x, y, z) = (1, -1, 2) \cdot (x, y, z) = x - y + 2z.$$

Then

$$W = \{v \in \mathbb{R}^3 \mid T(v) = 3\}.$$

Let

$$U = \ker(T) = \{v \in \mathbb{R}^3 \mid T(v) = 0\}.$$

Choose

$$v_0 = (3, 0, 0),$$

so that $T(v_0) = 3$. For $v \in W$,

$$T(v - v_0) = T(v) - T(v_0) = 3 - 3 = 0,$$

hence $v - v_0 \in U$. Therefore

$$W = v_0 + U.$$

So W is an affine translate of the subspace U , not a subspace itself.

$$\boxed{W \not\leq \mathbb{R}^3}$$

Method 3: Failure of closure under scalar multiplication

Take

$$u = (3, 0, 0) \in W,$$

since

$$3 - 0 + 0 = 3.$$

If W were a subspace, then $\frac{1}{2}u$ would also lie in W . But

$$\frac{1}{2}u = \left(\frac{3}{2}, 0, 0\right),$$

and

$$\frac{3}{2} - 0 + 0 = \frac{3}{2} \neq 3.$$

Thus W is not closed under scalar multiplication. Hence W is not a subspace.

$$\boxed{W \not\subseteq \mathbb{R}^3}$$

Question 13 (Polynomials Divisible by $(x - 1)^2$) (10 marks)

Problem

Let

$$W = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(1) = 0, p'(1) = 0\}.$$

Determine whether W is a subspace of $\mathcal{P}_4(\mathbb{R})$.

Solution

Method 1: Kernel of a linear map

Define

$$T : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad T(p) = (p(1), p'(1)).$$

The map $p \mapsto p(1)$ is linear, since evaluation at a fixed point defines a linear functional on $\mathcal{P}_4(\mathbb{R})$. Also, the map $p \mapsto p'(1)$ is linear, because differentiation is a linear map from $\mathcal{P}_4(\mathbb{R})$ to $\mathcal{P}_3(\mathbb{R})$, and evaluation at the fixed point $x = 1$ is linear on $\mathcal{P}_3(\mathbb{R})$. Therefore T is linear.

Now

$$p \in W \iff p(1) = 0 \text{ and } p'(1) = 0 \iff T(p) = (0, 0) \iff p \in \ker(T).$$

Hence

$$W = \ker(T).$$

Since the kernel of a linear map is a subspace, W is a subspace of $\mathcal{P}_4(\mathbb{R})$.

$$\boxed{W \leq \mathcal{P}_4(\mathbb{R})}$$

Method 2: Factorisation

Let $p \in W$. Since $p(1) = 0$, the polynomial $x - 1$ divides $p(x)$. Write

$$p(x) = (x - 1)q(x), \quad q(x) \in \mathcal{P}_3(\mathbb{R}).$$

Differentiate:

$$p'(x) = q(x) + (x - 1)q'(x).$$

Evaluating at $x = 1$, we get

$$p'(1) = q(1).$$

But $p'(1) = 0$, so $q(1) = 0$. Hence $x - 1$ also divides $q(x)$, and therefore

$$p(x) = (x - 1)^2 r(x)$$

for some $r(x) \in \mathcal{P}_2(\mathbb{R})$.

Thus

$$W = \{(x - 1)^2 r(x) \mid r \in \mathcal{P}_2(\mathbb{R})\} = \text{span}\{(x - 1)^2, x(x - 1)^2, x^2(x - 1)^2\}.$$

Therefore W is a subspace.

$$\boxed{W \leq \mathcal{P}_4(\mathbb{R})}$$

Method 3: Solve in coordinates

Let

$$p(x) = ax^4 + bx^3 + cx^2 + dx + e.$$

Then

$$p(1) = a + b + c + d + e,$$

and

$$p'(x) = 4ax^3 + 3bx^2 + 2cx + d, \quad p'(1) = 4a + 3b + 2c + d.$$

The conditions become

$$a + b + c + d + e = 0,$$

$$4a + 3b + 2c + d = 0.$$

Solve for d and e :

$$d = -4a - 3b - 2c,$$

$$e = 3a + 2b + c.$$

Hence

$$p(x) = a(x^4 - 4x + 3) + b(x^3 - 3x + 2) + c(x^2 - 2x + 1).$$

Therefore

$$W = \text{span}\{x^4 - 4x + 3, x^3 - 3x + 2, x^2 - 2x + 1\}.$$

So W is a subspace of $\mathcal{P}_4(\mathbb{R})$.

$$\boxed{W \leq \mathcal{P}_4(\mathbb{R})}$$

Question 14 (Intersection of Polynomial Subspaces) (10 marks)

Problem

Let

$$U = \{p(x) \in \mathcal{P}_3(\mathbb{R}) \mid p(1) = 0\},$$

and

$$V = \{p(x) \in \mathcal{P}_3(\mathbb{R}) \mid p(-1) = 0\}.$$

Determine whether

$$W = U \cap V$$

is a subspace of $\mathcal{P}_3(\mathbb{R})$.

Solution

Method 1: Intersection property

Both U and V are subspaces of $\mathcal{P}_3(\mathbb{R})$, since each is the kernel of an evaluation map:

$$U = \ker(p \mapsto p(1)), \quad V = \ker(p \mapsto p(-1)).$$

Therefore

$$W = U \cap V$$

is the intersection of two subspaces. Since the intersection of subspaces is always a subspace, W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

$$\boxed{W \leq \mathcal{P}_3(\mathbb{R})}$$

Method 2: Factorisation

If $p \in W$, then

$$p(1) = 0, \quad p(-1) = 0.$$

Hence both $x - 1$ and $x + 1$ divide $p(x)$. Since these factors are distinct,

$$(x - 1)(x + 1) = x^2 - 1$$

divides $p(x)$. Since $p \in \mathcal{P}_3(\mathbb{R})$, there exist $a, b \in \mathbb{R}$ such that

$$p(x) = (x^2 - 1)(ax + b).$$

Thus

$$W = \text{span}\{x(x^2 - 1), x^2 - 1\}.$$

Therefore W is a subspace.

$$\boxed{W \leq \mathcal{P}_3(\mathbb{R})}$$

Method 3: Kernel of a vector-valued map

Define

$$T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad T(p) = (p(1), p(-1)).$$

This map is linear. Moreover,

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

$$\boxed{W \leq \mathcal{P}_3(\mathbb{R})}$$

Question 15 (Image of a Linear Map)

(10 marks)

Problem

Let

$$W = \{(a + b, a - b, 2a) \in \mathbb{R}^3 \mid a, b \in \mathbb{R}\}.$$

Determine whether W is a subspace of $\mathbb{R}^3$.

Solution

Method 1: Image of a linear map

Define

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3, \quad T(a, b) = (a + b, a - b, 2a).$$

This map is linear, since each coordinate is a linear expression in a and b . By definition,

$$W = \text{im}(T).$$

Since the image of a linear map is a subspace, W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Method 2: Span description

We can rewrite

$$(a + b, a - b, 2a) = a(1, 1, 2) + b(1, -1, 0).$$

Hence

$$W = \text{span}\{(1, 1, 2), (1, -1, 0)\}.$$

Since a span is a subspace, W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Method 3: Linear combination test

Let

$$u = (a_1 + b_1, a_1 - b_1, 2a_1) \in W, \quad v = (a_2 + b_2, a_2 - b_2, 2a_2) \in W.$$

For $\lambda, \mu \in \mathbb{R}$,

$$\lambda u + \mu v = ((\lambda a_1 + \mu a_2) + (\lambda b_1 + \mu b_2), (\lambda a_1 + \mu a_2) - (\lambda b_1 + \mu b_2), 2(\lambda a_1 + \mu a_2)).$$

Let

$$a = \lambda a_1 + \mu a_2, \quad b = \lambda b_1 + \mu b_2.$$

Then

$$\lambda u + \mu v = (a + b, a - b, 2a) \in W.$$

Hence W is closed under linear combinations, so W is a subspace of $\mathbb{R}^3$.

$$\boxed{W \leq \mathbb{R}^3}$$

Question 16 (Matrices with Equal Row and Column Sums) (12 marks)

Problem

Let

$$W = \left\{ A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid a + b = c + d, a + c = b + d \right\}.$$

Determine whether W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

Solution

Method 1: Kernel of a linear map

Define

$$T : M_{2 \times 2}(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad T\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = ((a + b) - (c + d), (a + c) - (b + d)).$$

Each component is a linear expression in the entries of the matrix, so T is linear. Observe that

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

$W \leq M_{2 \times 2}(\mathbb{R})$

Method 2: Solve the linear constraints

The defining equations are

$$a + b = c + d, \quad a + c = b + d.$$

Subtracting the second from the first gives

$$b - c = c - b,$$

so

$$b = c.$$

Substituting $c = b$ into $a + b = c + d$ gives

$$a + b = b + d,$$

hence

$$a = d.$$

Therefore every matrix in W has the form

$$\begin{pmatrix} a & b \\ b & a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Thus

$$W = \text{span} \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\},$$

so W is a subspace.

$W \leq M_{2 \times 2}(\mathbb{R})$

Method 3: Intersection of two subspaces

Let

$$U = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid a + b = c + d \right\},$$

and

$$V = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) \mid a + c = b + d \right\}.$$

Each of U and V is the solution set of a homogeneous linear equation in the entries of the matrix, so each is a subspace of $M_{2 \times 2}(\mathbb{R})$. Moreover,

$$W = U \cap V.$$

Since the intersection of subspaces is always a subspace, W is a subspace of $M_{2 \times 2}(\mathbb{R})$.

$$\boxed{W \leq M_{2 \times 2}(\mathbb{R})}$$

Question 17 (Affine Constraint in a Polynomial Space) (12 marks)

Problem

Let

$$W = \{p(x) \in \mathcal{P}_3(\mathbb{R}) \mid p(1) + p'(1) = 1\}.$$

Determine whether W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

Solution

Method 1: Zero polynomial test

If W were a subspace, then it must contain the zero polynomial. However, for $p(x) = 0$,

$$p(1) + p'(1) = 0 + 0 = 0 \neq 1.$$

Thus $0 \notin W$. Therefore W is not a subspace of $\mathcal{P}_3(\mathbb{R})$.

$$\boxed{W \not\leq \mathcal{P}_3(\mathbb{R})}$$

Method 2: Affine translate of a kernel

Define

$$T : \mathcal{P}_3(\mathbb{R}) \rightarrow \mathbb{R}, \quad T(p) = p(1) + p'(1).$$

The map $p \mapsto p(1)$ is linear, since evaluation at a fixed point is linear. The map $p \mapsto p'(1)$ is also linear, since differentiation is linear and evaluation at a fixed point is linear. Hence T is linear.

Then

$$W = \{p \in \mathcal{P}_3(\mathbb{R}) \mid T(p) = 1\}.$$

Let

$$U = \ker(T) = \{p \in \mathcal{P}_3(\mathbb{R}) \mid T(p) = 0\}.$$

Choose $p_0(x) = 1$. Then

$$T(p_0) = p_0(1) + p'_0(1) = 1 + 0 = 1.$$

Hence for any $p \in W$,

$$T(p - p_0) = T(p) - T(p_0) = 1 - 1 = 0,$$

so $p - p_0 \in U$. Therefore

$$W = p_0 + U.$$

Thus W is an affine translate of the subspace U , not a subspace itself.

$$\boxed{W \not\leq \mathcal{P}_3(\mathbb{R})}$$

Method 3: Failure of closure under scalar multiplication

Take

$$p(x) = 1 \in W,$$

since

$$p(1) + p'(1) = 1 + 0 = 1.$$

If W were a subspace, then $2p(x) = 2$ would also belong to W . But

$$(2p)(1) + (2p)'(1) = 2 + 0 = 2 \neq 1.$$

Thus W is not closed under scalar multiplication, so it is not a subspace.

$$\boxed{W \not\subseteq \mathcal{P}_3(\mathbb{R})}$$

Question 18 (Polynomials with Equal Endpoint Values and Derivatives) (12 marks)

Problem

Let

$$W = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(1) = p(-1), p'(1) = p'(-1)\}.$$

Determine whether W is a subspace of $\mathcal{P}_4(\mathbb{R})$.

Solution

Method 1: Kernel of a linear map

Define

$$T : \mathcal{P}_4(\mathbb{R}) \rightarrow \mathbb{R}^2, \quad T(p) = (p(1) - p(-1), p'(1) - p'(-1)).$$

The map $p \mapsto p(1) - p(-1)$ is linear because it is the difference of two evaluation maps, each of which is linear. Similarly, $p \mapsto p'(1) - p'(-1)$ is linear because differentiation is linear and evaluation at fixed points is linear. Hence T is linear.

Now

$$p \in W \iff p(1) - p(-1) = 0, p'(1) - p'(-1) = 0 \iff T(p) = (0, 0).$$

Thus

$$W = \ker(T).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $\mathcal{P}_4(\mathbb{R})$.

$W \leq \mathcal{P}_4(\mathbb{R})$

Method 2: Coefficient comparison

Let

$$p(x) = ax^4 + bx^3 + cx^2 + dx + e.$$

Then

$$p(1) = a + b + c + d + e, \quad p(-1) = a - b + c - d + e.$$

Hence

$$p(1) = p(-1) \iff 2b + 2d = 0 \iff b + d = 0.$$

Also,

$$p'(x) = 4ax^3 + 3bx^2 + 2cx + d,$$

so

$$p'(1) = 4a + 3b + 2c + d, \quad p'(-1) = -4a + 3b - 2c + d.$$

Thus

$$p'(1) = p'(-1) \iff 8a + 4c = 0 \iff 2a + c = 0.$$

Therefore

$$d = -b, \quad c = -2a.$$

So

$$p(x) = a(x^4 - 2x^2) + b(x^3 - x) + e.$$

Hence

$$W = \text{span}\{x^4 - 2x^2, x^3 - x, 1\}.$$

Therefore W is a subspace.

$$\boxed{W \leq \mathcal{P}_4(\mathbb{R})}$$

Method 3: Even–odd decomposition

Every polynomial $p \in \mathcal{P}_4(\mathbb{R})$ can be written uniquely as

$$p(x) = p_e(x) + p_o(x),$$

where p_e is even and p_o is odd.

For the first condition,

$$p(1) = p(-1)$$

means that the odd part contributes zero at $x = 1$, since

$$p_o(-1) = -p_o(1).$$

Hence this condition is equivalent to

$$p_o(1) = 0.$$

For the second condition,

$$p'(1) = p'(-1).$$

Now the derivative of an even polynomial is odd, while the derivative of an odd polynomial is even. Thus this condition forces the odd part of p' to vanish at 1, equivalently

$$p'_e(1) = 0.$$

Write

$$p_e(x) = ax^4 + cx^2 + e, \quad p_o(x) = bx^3 + dx.$$

Then

$$p_o(1) = b + d = 0, \quad p'_e(x) = 4ax^3 + 2cx, \quad p'_e(1) = 4a + 2c = 0.$$

Thus

$$d = -b, \quad c = -2a,$$

so again

$$p(x) = a(x^4 - 2x^2) + b(x^3 - x) + e.$$

Hence

$$W = \text{span}\{x^4 - 2x^2, x^3 - x, 1\},$$

which is a subspace.

$$\boxed{W \leq \mathcal{P}_4(\mathbb{R})}$$

Question 19 (Smallest Subspace Containing a Parametric Family) (12 marks)

Problem

Let

$$S = \{(1, t, t^2) \in \mathbb{R}^3 \mid t \in \mathbb{R}\}.$$

Let W be the smallest subspace of $\mathbb{R}^3$ containing S .

Determine W .

Solution

Method 1: Produce three independent vectors from the family

Since $S \subseteq W$, the vectors corresponding to $t = 0, 1, -1$ all lie in W :

$$(1, 0, 0), \quad (1, 1, 1), \quad (1, -1, 1).$$

Now

$$(1, 1, 1) - (1, -1, 1) = (0, 2, 0),$$

so

$$(0, 1, 0) \in W.$$

Also,

$$(1, 1, 1) - (1, 0, 0) = (0, 1, 1),$$

hence

$$(0, 1, 1) - (0, 1, 0) = (0, 0, 1) \in W.$$

Therefore W contains

$$(1, 0, 0), \quad (0, 1, 0), \quad (0, 0, 1),$$

so $W = \mathbb{R}^3$.

$$\boxed{W = \mathbb{R}^3}$$

Method 2: Span containment both ways

For any $t \in \mathbb{R}$,

$$(1, t, t^2) = (1, 0, 0) + t(0, 1, 0) + t^2(0, 0, 1).$$

Thus

$$S \subseteq \text{span}\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\} = \mathbb{R}^3.$$

So the smallest subspace containing S satisfies

$$W \subseteq \mathbb{R}^3.$$

Conversely, as shown above, the three standard basis vectors are in the span of elements of S , so

$$\mathbb{R}^3 \subseteq W.$$

Hence

$$W = \mathbb{R}^3.$$

$$\boxed{W = \mathbb{R}^3}$$

Method 3: Use a contradiction via a nonzero linear functional

Suppose $W \neq \mathbb{R}^3$. Then W is a proper subspace of $\mathbb{R}^3$, so there exists a nonzero linear functional

$$L(x, y, z) = ax + by + cz$$

such that

$$W \subseteq \ker(L).$$

Since $S \subseteq W$, we must have

$$L(1, t, t^2) = a + bt + ct^2 = 0 \quad \text{for all } t \in \mathbb{R}.$$

But a polynomial that vanishes for all $t \in \mathbb{R}$ must be the zero polynomial. Hence

$$a = b = c = 0,$$

contradicting that L is nonzero.

Therefore $W = \mathbb{R}^3$.

$$\boxed{W = \mathbb{R}^3}$$

Question 20 (Range of the Differentiation Operator) (12 marks)

Problem

Let

$$W = \{p'(x) \mid p(x) \in \mathcal{P}_4(\mathbb{R}), p(1) = 0\} \subseteq \mathcal{P}_3(\mathbb{R}).$$

Determine whether W is a subspace of $\mathcal{P}_3(\mathbb{R})$, and determine W explicitly.

Solution

Method 1: Image of a linear map

Let

$$U = \{p(x) \in \mathcal{P}_4(\mathbb{R}) \mid p(1) = 0\}.$$

Since U is the kernel of the evaluation map $p \mapsto p(1)$, it is a subspace of $\mathcal{P}_4(\mathbb{R})$.

Now define

$$D : U \rightarrow \mathcal{P}_3(\mathbb{R}), \quad D(p) = p'.$$

Differentiation is linear, so D is linear. By definition,

$$W = \text{im}(D).$$

Since the image of a linear map is a subspace, W is a subspace of $\mathcal{P}_3(\mathbb{R})$.

To determine W , let

$$q(x) \in \mathcal{P}_3(\mathbb{R}).$$

Define

$$p(x) = \int_1^x q(t) dt.$$

Then $p \in \mathcal{P}_4(\mathbb{R})$, $p(1) = 0$, and

$$p'(x) = q(x).$$

So every $q \in \mathcal{P}_3(\mathbb{R})$ lies in W . Hence

$$W = \mathcal{P}_3(\mathbb{R}).$$

Therefore

$$\boxed{W = \mathcal{P}_3(\mathbb{R})}.$$

Method 2: Construct preimages of the standard basis

We show that the standard basis of $\mathcal{P}_3(\mathbb{R})$ lies in W .

First,

$$1 = (x - 1)'$$

Since $x - 1 \in \mathcal{P}_4(\mathbb{R})$ and $(x - 1)(1) = 0$, we have $1 \in W$.

Next,

$$x = \left(\frac{x^2 - 1}{2} \right)',$$

and $\frac{x^2-1}{2} \in \mathcal{P}_4(\mathbb{R})$ vanishes at $x = 1$, so $x \in W$.

Also,

$$x^2 = \left(\frac{x^3 - 1}{3} \right)',$$

and $\frac{x^3-1}{3}(1) = 0$, so $x^2 \in W$.

Similarly,

$$x^3 = \left(\frac{x^4 - 1}{4} \right)',$$

and $\frac{x^4-1}{4}(1) = 0$, so $x^3 \in W$.

Thus

$$1, x, x^2, x^3 \in W.$$

Hence

$$\mathcal{P}_3(\mathbb{R}) \subseteq W.$$

Since $W \subseteq \mathcal{P}_3(\mathbb{R})$ by definition, it follows that

$$W = \mathcal{P}_3(\mathbb{R}).$$

Therefore W is a subspace and

$$\boxed{W = \mathcal{P}_3(\mathbb{R})}.$$

Method 3: Coefficient chase

Let

$$p(x) = ax^4 + bx^3 + cx^2 + dx + e \in \mathcal{P}_4(\mathbb{R}),$$

with

$$p(1) = a + b + c + d + e = 0.$$

Then

$$p'(x) = 4ax^3 + 3bx^2 + 2cx + d.$$

So every element of W has the form

$$4ax^3 + 3bx^2 + 2cx + d$$

for some $a, b, c, d, e \in \mathbb{R}$ satisfying

$$a + b + c + d + e = 0.$$

But for arbitrary $a, b, c, d \in \mathbb{R}$, we may choose

$$e = -(a + b + c + d),$$

so the condition $p(1) = 0$ imposes no restriction on the derivative.

Hence every polynomial of the form

$$\alpha x^3 + \beta x^2 + \gamma x + \delta$$

can occur as $p'(x)$ for some $p \in \mathcal{P}_4(\mathbb{R})$ with $p(1) = 0$. Therefore

$$W = \mathcal{P}_3(\mathbb{R}).$$

So W is a subspace, and

$$\boxed{W = \mathcal{P}_3(\mathbb{R})}.$$

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Question 21 (Characteristic-Polynomial Constraint) (12 marks)

Problem

Let

$$W = \{A \in M_{n \times n}(\mathbb{R}) \mid \chi_A(A) = 0\},$$

where $\chi_A(\lambda) = \det(\lambda I - A)$ is the characteristic polynomial of A .

Determine whether W is a subspace of $M_{n \times n}(\mathbb{R})$.

Solution

Method 1: Cayley–Hamilton theorem

For every $A \in M_{n \times n}(\mathbb{R})$, the Cayley–Hamilton theorem states that

$$\chi_A(A) = 0.$$

Hence every matrix $A \in M_{n \times n}(\mathbb{R})$ belongs to W . Therefore

$$W = M_{n \times n}(\mathbb{R}).$$

Since the whole ambient space is a subspace of itself, W is a subspace.

$$\boxed{W \leq M_{n \times n}(\mathbb{R})}$$

Method 2: Identify the set immediately

The defining condition

$$\chi_A(A) = 0$$

is not a restriction at all, because it is satisfied by every square matrix. Thus

$$W = M_{n \times n}(\mathbb{R}),$$

and so W is a subspace.

$$\boxed{W \leq M_{n \times n}(\mathbb{R})}$$

Method 3: Check by reinterpretation

The statement $\chi_A(A) = 0$ says that A is annihilated by its own characteristic polynomial. By Cayley–Hamilton, this holds for every square matrix A . Hence the set W is exactly the whole space $M_{n \times n}(\mathbb{R})$, and therefore it is a subspace.

$$\boxed{W \leq M_{n \times n}(\mathbb{R})}$$

Question 22 (Operators Preserving a Fixed Subspace) (12 marks)

Problem

Let V be a finite-dimensional real vector space, and let $U \leq V$ be fixed.

Define

$$W = \{T \in \mathcal{L}(V) \mid T(U) \subseteq U\}.$$

Determine whether W is a subspace of $\mathcal{L}(V)$.

Solution

Method 1: Quotient-map kernel trick

Let

$$\iota : U \hookrightarrow V$$

be the inclusion map, and let

$$\pi : V \rightarrow V/U$$

be the quotient map.

For $T \in \mathcal{L}(V)$, consider the map

$$\Phi : \mathcal{L}(V) \rightarrow \mathcal{L}(U, V/U), \quad \Phi(T) = \pi \circ T \circ \iota.$$

Since composition of linear maps is linear in T , the map Φ is linear.

Now

$$\Phi(T) = 0 \iff \pi(T(u)) = 0 \text{ for all } u \in U \iff T(u) \in U \text{ for all } u \in U.$$

Thus

$$W = \ker(\Phi).$$

Since the kernel of a linear transformation is a subspace, W is a subspace of $\mathcal{L}(V)$.

$$\boxed{W \leq \mathcal{L}(V)}$$

Method 2: Block-matrix form relative to a decomposition

Choose a complementary subspace Z such that

$$V = U \oplus Z.$$

Relative to a basis adapted to this decomposition, every $T \in \mathcal{L}(V)$ has block form

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where $A : U \rightarrow U$, $B : Z \rightarrow U$, $C : U \rightarrow Z$, and $D : Z \rightarrow Z$.

The condition $T(U) \subseteq U$ is equivalent to saying that the Z -component of $T(u)$ is zero for all $u \in U$, that is,

$$C = 0.$$

Hence

$$W = \left\{ \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} \right\}.$$

This set is closed under linear combinations, so it is a subspace of $\mathcal{L}(V)$.

$$\boxed{W \leq \mathcal{L}(V)}$$

Method 3: Direct linear-combination test

Let $T_1, T_2 \in W$, and let $a, b \in \mathbb{R}$. For any $u \in U$,

$$(aT_1 + bT_2)(u) = aT_1(u) + bT_2(u).$$

Since $T_1(u) \in U$ and $T_2(u) \in U$, and since U is a subspace, it follows that

$$aT_1(u) + bT_2(u) \in U.$$

Hence $(aT_1 + bT_2)(U) \subseteq U$, so $aT_1 + bT_2 \in W$. Therefore W is a subspace of $\mathcal{L}(V)$.

$$\boxed{W \leq \mathcal{L}(V)}$$

Question 23 (Operators Acting as the Identity on a Fixed Subspace) (12 marks)

Problem

Let V be a finite-dimensional real vector space, and let $U \leq V$ be fixed.

Define

$$W = \{T \in \mathcal{L}(V) \mid T(u) = u \text{ for all } u \in U\}.$$

Determine whether W is a subspace of $\mathcal{L}(V)$.

Solution

Method 1: Affine translate of a kernel

Define

$$\Psi : \mathcal{L}(V) \rightarrow \mathcal{L}(U, V), \quad \Psi(T) = T|_U.$$

Restriction to U is linear, so Ψ is linear.

Let

$$I_U : U \rightarrow V, \quad I_U(u) = u.$$

Then

$$W = \{T \in \mathcal{L}(V) \mid \Psi(T) = I_U\}.$$

Thus W is not the kernel of a linear map, but the preimage of a nonzero vector under a linear map. Equivalently, if $T_0 = I_V$, then

$$W = T_0 + \ker(\Psi).$$

So W is an affine translate of the subspace $\ker(\Psi)$. Since it is translated by the nonzero operator I_V , it is generally not a subspace.

Indeed, $0 \notin W$, since $0(u) = 0 \neq u$ for nonzero $u \in U$. Therefore W is not a subspace.

$$\boxed{W \not\leq \mathcal{L}(V)}$$

Method 2: Zero operator test

If W were a subspace, then it would contain the zero operator. However, the zero operator satisfies

$$0(u) = 0$$

for all $u \in U$, whereas the defining condition for W is

$$T(u) = u$$

for all $u \in U$.

If $U \neq \{0\}$, then these are not the same. Hence

$$0 \notin W.$$

Therefore W is not a subspace of $\mathcal{L}(V)$.

$$\boxed{W \not\leq \mathcal{L}(V)}$$

Method 3: Failure of closure under scalar multiplication

Let $T \in W$. Then $T(u) = u$ for all $u \in U$. If W were a subspace, then $\frac{1}{2}T \in W$ should also hold. But for $u \in U$,

$$\left(\frac{1}{2}T\right)(u) = \frac{1}{2}T(u) = \frac{1}{2}u \neq u$$

for $u \neq 0$.

Thus $\frac{1}{2}T \notin W$. Hence W is not closed under scalar multiplication, so it is not a subspace.

$$\boxed{W \not\subseteq \mathcal{L}(V)}$$

Question 24 (Zero Schur Complement Family) (12 marks)

Problem

Fix an invertible matrix $A_0 \in M_{n \times n}(\mathbb{R})$. Define

$$W = \left\{ \begin{pmatrix} 0 & B \\ C & D \end{pmatrix} \in M_{2n \times 2n}(\mathbb{R}) \mid D - CA_0^{-1}B = 0 \right\}.$$

Determine whether W is a subspace of $M_{2n \times 2n}(\mathbb{R})$.

Solution

Method 1: Schur-complement interpretation

For any blocks B, C, D , consider the block matrix

$$M = \begin{pmatrix} A_0 & B \\ C & D \end{pmatrix}.$$

Since A_0 is invertible, the Schur complement of A_0 in M is

$$S_{A_0} = D - CA_0^{-1}B.$$

Thus the defining condition for W is precisely that the Schur complement of A_0 be zero.

However, this condition is bilinear in B and C , so it is not expected to define a linear subspace. We now give a concrete counterexample.

Take

$$X = \begin{pmatrix} 0 & B \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ C & 0 \end{pmatrix},$$

with nonzero B, C . Then

$$0 - 0 \cdot A_0^{-1}B = 0, \quad 0 - CA_0^{-1}0 = 0,$$

so $X, Y \in W$.

But

$$X + Y = \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix},$$

and its defining condition would require

$$0 - CA_0^{-1}B = 0.$$

For suitable nonzero B, C , this fails. For example, take $B = C = I_n$. Then

$$CA_0^{-1}B = A_0^{-1} \neq 0.$$

Hence $X + Y \notin W$. Therefore W is not closed under addition, and so it is not a subspace.

$$\boxed{W \not\subseteq M_{2n \times 2n}(\mathbb{R})}$$

Method 2: Explicit counterexample with $B = C = I_n$

Let

$$X = \begin{pmatrix} 0 & I_n \\ 0 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & 0 \\ I_n & 0 \end{pmatrix}.$$

Then $X \in W$ because its blocks satisfy

$$D - CA_0^{-1}B = 0 - 0 \cdot A_0^{-1}I_n = 0.$$

Similarly,

$$Y \in W$$

because

$$0 - I_n A_0^{-1} 0 = 0.$$

But

$$X + Y = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix},$$

and here

$$D - CA_0^{-1}B = 0 - I_n A_0^{-1} I_n = -A_0^{-1} \neq 0.$$

So $X + Y \notin W$. Hence W is not a subspace.

$$\boxed{W \not\subseteq M_{2n \times 2n}(\mathbb{R})}$$

Method 3: Structural reason

The condition

$$D = CA_0^{-1}B$$

is linear in D , but bilinear in B and C . A set defined by such a bilinear relation is generally not stable under addition. The explicit matrices above show that this failure really occurs, so W is not a subspace.

$$\boxed{W \not\subseteq M_{2n \times 2n}(\mathbb{R})}$$

Question 25 (Smallest Subspace Containing All Powers of a Matrix) (12 marks)

Problem

Let $A \in M_{n \times n}(\mathbb{R})$ be fixed, and let W be the smallest subspace of $M_{n \times n}(\mathbb{R})$ containing

$$\{I, A, A^2, A^3, \dots\}.$$

Determine W .

Solution

Method 1: Cayley–Hamilton reduction

By the Cayley–Hamilton theorem, A satisfies its characteristic polynomial. Thus there exist scalars $c_0, c_1, \dots, c_{n-1}$ such that

$$A^n = c_{n-1}A^{n-1} + \dots + c_1A + c_0I.$$

Hence $A^n \in \text{span}\{I, A, \dots, A^{n-1}\}$.

Multiplying the above relation by A , we get

$$A^{n+1} \in \text{span}\{A, A^2, \dots, A^n\} \subseteq \text{span}\{I, A, \dots, A^{n-1}\}.$$

Repeating this argument inductively, we obtain

$$A^k \in \text{span}\{I, A, \dots, A^{n-1}\} \quad \text{for all } k \geq n.$$

Therefore every power of A lies in $\text{span}\{I, A, \dots, A^{n-1}\}$. Since W is the smallest subspace containing all powers of A , it follows that

$$W = \text{span}\{I, A, \dots, A^{n-1}\}.$$

Thus

$$\boxed{W = \text{span}\{I, A, \dots, A^{n-1}\}}.$$

Method 2: Image of a polynomial-evaluation map

Define

$$\Phi : \mathcal{P}(\mathbb{R}) \rightarrow M_{n \times n}(\mathbb{R}), \quad \Phi(p) = p(A).$$

This map is linear. The set of all powers of A is contained in $\text{im}(\Phi)$, and in fact

$$\text{im}(\Phi) = \text{span}\{I, A, A^2, \dots\}.$$

By Cayley–Hamilton, every polynomial $p(A)$ can be reduced modulo χ_A to a polynomial of degree at most $n - 1$. Hence

$$\text{im}(\Phi) = \text{span}\{I, A, \dots, A^{n-1}\}.$$

Therefore the smallest subspace containing all powers of A is

$$\boxed{W = \text{span}\{I, A, \dots, A^{n-1}\}}.$$

Method 3: Minimal-polynomial viewpoint

Let $m_A(x)$ be the minimal polynomial of A , and suppose

$$\deg(m_A) = r.$$

Then $m_A(A) = 0$, so A^r is a linear combination of $I, A, \dots, A^{r-1}$. It follows that every higher power of A is also a linear combination of $I, A, \dots, A^{r-1}$. Hence

$$W = \text{span}\{I, A, \dots, A^{r-1}\}.$$

In particular, since $r \leq n$, we certainly have

$$W = \text{span}\{I, A, \dots, A^{n-1}\}.$$

Thus

$$\boxed{W = \text{span}\{I, A, \dots, A^{n-1}\}}.$$